

We would like to begin by thanking both referees and the editor for their thoughtful consideration and insightful feedback. We pursued all recommended avenues and address each point below. In total, with the direction you provided, we feel we have a robust paper, based on our original metric.

Following your suggestion, we made a significant revision by replacing the 2-Stage Least-Squares Model with a Stock-Flow model. We feel the paper benefits substantially from this enhanced reframing.

#### **World Bank Data:**

##### **Your comment:**

*On page 2, still in the Introduction, the authors state, “For most of its history, the U.S. was less densely populated than its high-income peers, but in 1992 this changed. In that year, the U.S. grew to 28 people per square kilometer, outpacing the High Income Nations whose average was 23 people per square kilometer.” As you can see in the graph below, there was an extreme discontinuity in the High Income Nation data between 1991-1992.*

*In fact, in 1991, the density per square kilometer in High Income Nations was 34. I don’t know what happened to the data, but I don’t recall a global event of such significance that it would have decimated populations and/or destroyed enough land area in High Income Nations to result in the graph we see here. This gives me pause about using the World Bank data as your only source. I would encourage you to look for other data wherever you were previously relying solely on World Bank data. For what it is worth, the U.S. data from the World Bank does not exhibit this same pattern.*

##### **Our response:**

We pulled fresh data from WorldBank which did not exhibit the same discontinuity. We revised our statement to,

“Between 2000 and 2022, population density in high-income countries increased by approximately 12 percent, while density in the United States increased by roughly 19 percent.”

#### **Early Definition:**

##### **Your comment:**

*Also on page 2, you define your Rental Density Index as “the quotient of population over rental units.” I almost did not read any further. If that were actually the definition, that would be fraught with too many issues to warrant further study. Fortunately, I pushed on, but it wasn’t really until page 7 that I got a better, cleaner definition. You need to make sure the reader gets to that information sooner, and do not muddy the proverbial waters with a loosely-worded definition that offers an incomplete/erroneous picture of what you are doing.*

**Our response:**

We are grateful that you continued reading despite the poorly worded definition. We now introduce the proper definition in the abstract and reiterate it in the introduction before formally explaining the mathematics in Data and Descriptive Statistics.

**Causes of RDI Movement:****Your comment:**

*Still in the Introduction on page 4, the authors state, “The reasons for an increasing RDI are, in most cases, population growth in excess of supply growth. However, it could be possible that housing units declined as populations remained static.” You need to identify any cases where that happened in your data, pull them out, and re-run your tests. Whether you fully report the results or just reference them in a footnote, you need to make sure it is as rare as you think and that it does not impact your results.*

**Our response:**

Our statement was indeed reductive, and we should have fleshed it out. We revised with the following:

Although changes in RDI can arise mechanically through changes in either renter population or renter households, declines in rental unit counts are rare in our sample, occurring in fewer than 50 of 2,000 CBSA-year observations. And we explicitly control for supply.

**Explanation of RDI and its delta:****Your Comment:**

*Related to this discussion, the general idea you offer here is that RDI increases point to a greater latent rental demand but this behavior results in lower rents. Into a land of lower rents walks a multifamily developer to build an apartment complex. That makes little sense, and I would argue that it flies in the face of the Wheaton-DiPasquale Four-Quadrant model. I think some additional attention needs to be paid to the story/explanation here.*

**Our response:**

We revised our framing based on this comment, shifting to a Wheaton-DiPasquale framework.

**Consideration of MSA-level Inflation****Your Comment:**

*Is there a way to come up with MSA-level inflation? I know you mention this in Section 3, but it would really help the analysis. Is there any way to back this out of any data the FHFA makes available?*

**Our Response:**

We carefully considered the feasibility of constructing MSA-level inflation adjustments. We explored both FHFA metropolitan house price indexes and HUD Fair Market Rent series as potential candidates. FHFA indexes measure transaction prices in the owner-occupied housing market and are dominated by single-family sales, making them conceptually mismatched to rental rent growth in multifamily markets. HUD Fair Market Rents, while rental-based, are policy benchmarks designed for voucher administration; they incorporate smoothing, lagged information, and national CPI inputs and are not intended to measure year-to-year market rent inflation. Because neither source provides a consistent, market-based measure of local rental inflation over our sample period, we instead remove common inflationary and housing-cycle effects using year fixed effects and by normalizing rent growth relative to the national median each year. We clarify this choice in the revised manuscript.

**2SLS Reconsideration:****Your Comments:**

*Section 4.1, Page 12: "Individuals typically begin to form independent rental households in their early 20s. Therefore, the size of the birth cohort from 20 years ago directly influences the intensity of household formation pressure today." Historically, this is absolutely true. Over the time period of your analysis, this is less- and-less the case every year. For this reason, I am not sold on the use of birth cohort size as a valid instrument. I also think that the current share of individuals born in the MSA 20 years prior is absolutely tied to concurrent housing affordability issues. If rents recently spiked, that is most assuredly going to lead to an outmigration of 20-somethings. Are you controlling for rate of higher education attendance across MSAs? That will also impact your measured variable here.*

*What happens if the propensity to form independent households changes over time? It's been documented at length in the ownership literature that young people who lived through the Great Financial Crisis delayed housing purchases. What if there is a similar behavioral mechanism at work with rental household formation? Has anyone looked at that idea? I don't think it is a far-fetched concern that lingering effects from the GFC could be depressing the rate of new rental household formation. Perhaps look at a subset of cities hardest hit by the subprime mortgage fallout.*

*Two of your robustness checks are somewhat questionable, in my opinion. The placebo test that you report is essentially a tautology. In other words, it is of course the case that rents are going to respond to lagged RDI rather than contemporaneous RDI. Residential rents are set once per year, typically. Therefore, it will take a non-trivial amount of time for rents to catch up to market conditions. In a similar vein, I am not sure what the purpose of running a version of the instrument lagged by one year.*

**Our Response:**

We replaced the 2SLS model with a stock-flow model per your suggestion.

**Data Question on Vacant Dwellings:****Your Comment:**

*Page 7 in the Data and Descriptive Statistics section: where are you getting data on vacant units and what is a seasonal dwelling in this context? Also, where does an AirBNB fall?*

**Our Response:**

The IPUMS data include both occupied and vacant housing units, with vacancy status identified by the VACANCY variable. Our measures of renter household density and bedroom density are constructed using only renter-occupied households, and therefore exclude vacant and seasonal units by construction. An Airbnb unit would only be included if its owner reported it as their primary dwelling. We clarified this in the manuscript

**Large Home Builders' Perspective:****Your Comment:**

*Part of the setup is an argument that “...when all available and affordable units are occupied, latent demand becomes invisible to market participants and researchers alike.” I disagree with that supposition. Multifamily developers routinely see through that cloak of invisibility. It would add a layer of richness to this study to talk to one or more large multifamily developers about how they analyze these same questions you are trying to address. They may have some insight that would really ratchet up the contribution of this piece. Alternatively, they may be impressed by the thoughtful and applied research you are doing that could directly impact their decision-making and bottom line. Either way, that is a win for the authors, in my view.*

**Our Response:**

This comment was well taken, and we reached out to the homebuilding unit at Brookfield. Based on a discussion with their team, we bolstered the background section with reference to their methods and outlook.

**Estimates of Housing Shortage:****Your Comment:**

Your sentence in the first paragraph of the Introduction about the size of the housing unit shortfall in the US (4.4 million housing units). There is a good bit of debate about that number, so you might want to provide a range of estimates from a couple of reputable sources.

**Our Response:**

We agree and included a chart in the appendix, outlining recent estimates of housing shortage, noting their variance across time and magnitude.

**Formatting Revisions:**

**References:** your comments noted our LaTeX program's glitch in formatting our references.

We revised these references

**Diction:** your comments noted our 'use of "outperform" and "underperform" when discussing rent increases. It is not entirely clear to me what those terms mean in this context.'

We define these at first use.

**Map Graphics:** your comment referenced the difficulty in reading the maps we used

We agree and removed the maps.

**Writing Quality:**

**Your comments** referenced uncontextualized statements and poorly worded summaries of prior research. We performed a thorough review and rewrite.